

COURSE PLAN

CRYPTOGRAPHY & NETWORK SECURITY LAB

Prepared by

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UNIT-WISE COURSE PLAN

CRYPTOGRAPHY & NETWORK SECURITY LAB

Total Hours: 45 (15 sessions × 3 hours)

Software / Tools: GCC, JDK (javax.crypto, keytool), web browser, Wireshark, tcpdump / WinDump, Nmap, Snort, arpspoof / Ettercap; Linux or Windows

Reference Books: (1) William Stallings, *Computer Security – Principles and Practices*, 2/e, Pearson Education; (2) William Stallings, *Cryptography and Network Security*, Pearson Education Asia, New Delhi

UNIT I: Bitwise Operations and Classical Ciphers — 9 Hours

S.No.	Experiment / Topic	Hrs	Mode of Teaching	Learning Activity
1	Lab introduction and ethics of security testing; C programs to XOR each character of “Hello world” with 0, and to AND, OR and XOR each character with 127	3	Lecture / Demonstration	Observe and explain the effect of each bitwise operation on the output
2	Java programs for encryption and decryption using the Caesar cipher and the Substitution cipher	3	Demonstration / Guided Practice	Encrypt and decrypt messages; attempt a brute-force attack on Caesar
3	Java program for encryption and decryption using the Hill cipher	3	Worked Examples / Hands-on Practice	Compute key-matrix inverse mod 26; verify decryption

UNIT II: Symmetric Block and Stream Ciphers — 12 Hours

S.No.	Experiment / Topic	Hrs	Mode of Teaching	Learning Activity
1	C/Java program to implement the DES algorithm logic	3	Demonstration / Guided Practice	Trace key generation and one Feistel round
2	C/Java program to implement the Blowfish algorithm logic	3	Guided Hands-on Practice	Implement subkey generation and the F-function
3	C/Java program to implement the Rijndael (AES) algorithm logic	3	Worked Examples / Hands-on Practice	Implement SubBytes, ShiftRows, MixColumns and AddRoundKey
4	RC4 logic in Java; encrypt “Hello world” with Blowfish using Java Cryptography and a key created with Java keytool	3	Demonstration / Hands-on Practice	Generate a key with keytool; use javax.crypto for encryption

UNIT III: Public-Key Cryptography and Message Digests — 9 Hours

S.No.	Experiment / Topic	Hrs	Mode of Teaching	Learning Activity
1	Java program to implement the RSA algorithm	3	Worked Examples / Hands-on Practice	Generate keys; encrypt and decrypt a message
2	Diffie-Hellman key exchange mechanism using HTML and JavaScript	3	Demonstration / Hands-on Practice	Build a web page showing both parties computing the shared key
3	Message digest of a text using SHA-1 and MD5 in Java	3	Guided Hands-on Practice	Compare digest lengths; observe the avalanche effect

UNIT IV: Network Security — 12 Hours

S.No.	Experiment / Topic	Hrs	Mode of Teaching	Learning Activity
1	Find the IP address and MAC address of your machine; find neighbouring machines in the network; check if a server is up and running	3	Demonstration / Guided Practice	Use ipconfig/ifconfig, arp, ping and nmap
2	Run the tcpdump / WinDump utility with at least 4 options	3	Guided Hands-on Practice	Capture and filter traffic by host, port and protocol
3	Capture packets using Wireshark and analyse one TCP packet in detail	3	Demonstration / Hands-on Practice	Analyse TCP header fields and the three-way handshake
4	Use Snort to detect intrusion packets; demonstrate ARP poisoning on an isolated lab network	3	Case Study / Hands-on Practice	Write Snort rules and view alerts; observe poisoned ARP tables

UNIT V: Revision and Assessment — 3 Hours

S.No.	Experiment / Topic	Hrs	Mode of Teaching	Learning Activity
1	Revision, Record Evaluation and Internal Lab Examination	3	Viva-voce / Practical Test	Program execution, record submission and viva

Overall Hour Distribution

Unit	Unit Title	Hours	Percentage
I	Bitwise Operations and Classical Ciphers	9	20.00%
II	Symmetric Block and Stream Ciphers	12	26.67%
III	Public-Key Cryptography and Message Digests	9	20.00%
IV	Network Security	12	26.67%
V	Revision and Assessment	3	6.67%
	GRAND TOTAL	45	100%

General Modes of Teaching

- Each laboratory session is of 3 hours: a short briefing/demonstration by the faculty, followed by individual hands-on implementation by students.
- Students prepare the algorithm/design before the session, and execute, test and debug the programs during the session.
- Cipher outputs are verified by hand on small inputs and against standard library implementations (javax.crypto, MessageDigest).
- Network security experiments are carried out only on the isolated lab network, with emphasis on ethical and legal use of the tools.
- Guided problem solving for complex experiments; additional exercises are given to fast learners.
- Continuous assessment through observation book, lab record, program output and viva-voce in every session.
- The final session is for revision, record evaluation and the internal laboratory examination.